

Department of Mathematics and Applied Mathematics
Module 3MS: Metric Spaces

Class Test Two

26 April 2016

Time : 18h00 – 19h30

Total marks: 40

1. (a) Let d_D be the discrete metric on $\mathbb{R}$. What are the Cauchy sequences in this space? No proof required.
(b) Let d_D be the discrete metric on $\mathbb{R}$. Is this space complete? Only an answer, no proof, required.
(c) Now let d_E be the Euclidean metric. What is the completion of $((0, 1], d_E)$?
No proof required. [3]

2. Prove that every complete subset of a metric space is closed. [5]

3. (a) Prove that every Lipschitz-continuous function is continuous.
(b) Write down two necessary and sufficient conditions for a function between metric spaces to be continuous, which are different to the one you used in (a) above. No proofs required.
(c) Let ρ be the metric on $[0, \infty)$ defined by $\rho(x, y) = |x^2 - y^2|$.
Let $f : ([0, \infty), d_E) \rightarrow ([0, \infty), \rho)$ be given by $f(x) = x$. Is f uniformly continuous? Justify your answer.
(d) Recall that we say P is a “topological property” if, whenever a metric space X has property P , and the metric space Y is homeomorphic to X , then Y also has property P . Show that the property of having a finite diameter is *not* a topological property.

[15]

4. Read the theorem below, and answer the questions that follow.

Theorem If A and B are complete subsets of a metric space (X, d) , then $A \cup B$ is complete.

Proof: Suppose that (x_n) is a Cauchy sequence in $A \cup B$. (1)

Then (x_n) must have a subsequence in A or in B (or both). (2)

Suppose without loss of generality that (x_{n_k}) is a subsequence of (x_n) in A . (3)

Then (x_{n_k}) is a Cauchy sequence in A . (4)

So $x_{n_k} \rightarrow a$, for some $a \in A$. (5)

So $x_n \rightarrow a$. (6)

Since $a \in A \cup B$, this completes the proof. (7)

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- (a) Justify the claim in line (4).
- (b) Why do we know that (x_{n_k}) converges, as claimed in line (5)?
- (c) Why do we know that (x_n) converges, as claimed in line (6)?
- (d) Provide an example of a metric space (specify the set and the metric) and a collection of complete subsets of the space whose union is not complete. No proof required.
- (e) Is it true that the union of any collection of open, complete subsets of a metric space is complete? Justify your answer. [11]

5. (a) Let $\mathcal{T} = \{A \subseteq \mathbb{R} : 8 \notin A\} \cup \{\mathbb{R}\}$.

- i. Show that $\mathcal{T}$ is a topology on $\mathbb{R}$.
- ii. Show that the topology $\mathcal{T}$ is not induced by a metric.

(b) Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \rightarrow Y$ a function from X to Y .

For $x \in X$, define:

$$w_f(x) = \inf\{\text{diam } f(G) : G \text{ is an open subset of } X \text{ containing } x\}$$

What familiar property of the function f is given by the statement " $w_f(x) = 0$ "?

State clearly what the property is, and provide a proof to justify your answer.

[6]